

SIMATS ENGINEERING

CO5-ASSESSMENT TOOL3-Reverse Engineering

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO5: Design intelligent agents and real-time AI-based systems (chatbots, expert systems, emotion detection, edge AI, autonomous drones, and related applications) by selecting appropriate agent architectures, knowledge representation, and reasoning mechanisms to solve real-world problems. (BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Q1. The following is a conversation between a user and an AI-based flight-booking chatbot. Study the transcript given below and reverse-engineer the chatbot's design by: (a) writing its PEAS description, (b) classifying the type of agent it represents with justification, (c) identifying the likely component (NLU, Dialogue Manager, Knowledge Base, or NLG) responsible for each chatbot response, and (d) suggesting one enhancement using a learning-agent architecture.

Speaker	Message
User	I want to book a flight to Chennai next Friday.
Chatbot	Sure! What time would you like to depart, and do you have a preferred airline?
User	Morning flight, no preference.
Chatbot	I found 3 morning flights to Chennai on Friday. Shall I show you the cheapest option first?

Q2. The block diagram below shows the processing pipeline of a Real-Time Emotion Detection System. Reverse-engineer the system by: (a) explaining the function performed at each stage of the pipeline, (b) identifying the likely AI/ML technique used at each stage, (c) writing the PEAS description of the system, and (d) classifying the

Real-Time Emotion Detection System — processing pipeline



agent type (e.g., simple reflex, model-based) with justification.

Q3. An Expert System Shell for medical diagnosis contains the rule base, input facts, and reasoning trace given below. Study it and reverse-engineer the shell by: (a) identifying the type of chaining (forward/backward) used to reach the diagnosis, with justification, (b) identifying the expert-system-shell components illustrated (knowledge base, inference engine, explanation facility), (c) explaining how the shell would perform knowledge acquisition to add a new rule for dengue, and (d) explaining, with reasoning, whether backward chaining could reach the same diagnosis.

Rule	IF (condition)	THEN (conclusion)
R1	fever = yes AND cough = yes	suspect = flu
R2	suspect = flu AND body_ache = yes	diagnosis = Influenza
R3	fever = yes AND rash = yes	suspect = measles

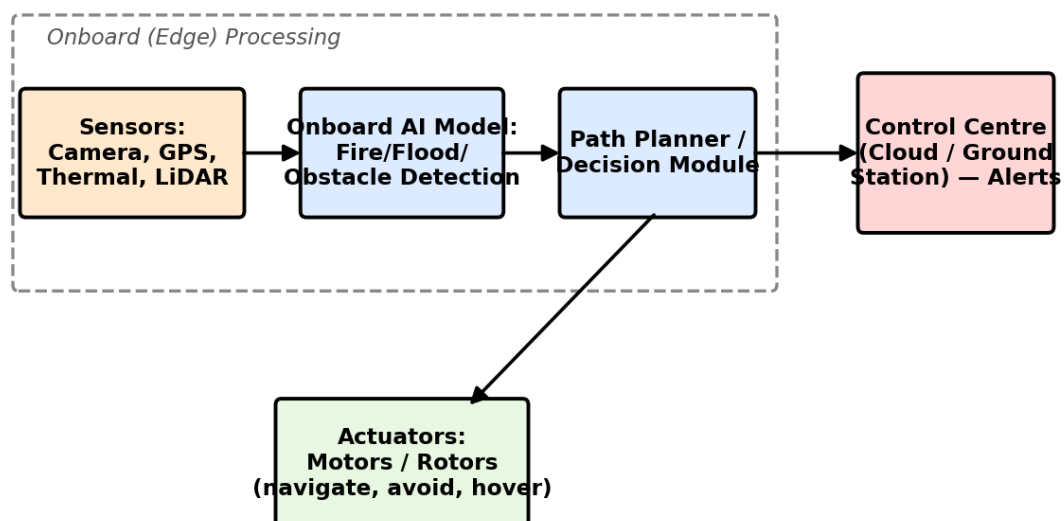
Given facts: fever = yes, cough = yes, body_ache = yes

Reasoning trace: R1 fires first (fever & cough matched) → suspect = flu. R2 then fires (suspect = flu & body_ache matched) → diagnosis = Influenza.

Explanation generated by the shell: “Diagnosis = Influenza because fever and cough indicated flu, and flu together with body ache confirmed Influenza.”

Q4. The diagram below shows the system architecture of an Autonomous Drone used for Disaster Detection with Edge AI deployment. Reverse-engineer the design by: (a) writing the PEAS description of the drone agent, (b) classifying the agent type (goal-based/utility-based) with justification, (c) identifying which components run on the edge versus the cloud and justifying why on-edge processing was chosen for detection and obstacle avoidance, and (d) identifying whether the agent's control architecture is reactive, deliberative, or hybrid, with justification.

Autonomous Drone — Disaster Detection (Edge AI) system architecture



Q5. The table below shows sample output logs from an AI-based Deepfake & Face Authenticity Verification system across five video frames. Study the output and reverse-engineer the system by: (a) identifying the likely input features used by the model (e.g., blink rate, texture/artifact score), (b) identifying the type of model/technique likely used to produce these outputs, (c) explaining how the final Real/Fake decision may have been derived from the frame-level confidence scores,

and (d) suggesting one way a swarm/ensemble-based approach could improve the system's reliability.

Frame No.	Blink Rate (per 10s)	Texture/Artifact Score	Fake Confidence (%)	Prediction
1	4	0.12	8	Real
2	1	0.61	74	Fake
3	0	0.83	91	Fake
4	5	0.09	5	Real
5	2	0.55	68	Fake

Overall system verdict: Fake (3 of 5 frames flagged — majority vote)

Code of Conduct:

I _____Venu G(192424390)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Identification of Agent Type & PEAS (correctly inferred from the given artifact)	Correctly identifies the agent type and gives a complete, accurate PEAS description, fully consistent with the given artifact.	Identifies the agent type and PEAS description correctly with only minor omissions.	Partially correct identification of agent type/PEAS; some elements are missing or inconsistent with the artifact.	Agent type/PEAS identification is largely incorrect or not attempted.	10
Identification of Architecture & Components (mapping artifact evidence to system components)	Accurately identifies all relevant architectural components and correctly maps each to specific evidence in the artifact.	Identifies most components correctly with reasonable mapping to the artifact; minor gaps present.	Identifies some components; mapping to the artifact is weak or partially incorrect.	Little or no correct identification of architecture/components.	10
Identification of Underlying Technique/Algorithm/Reas	Correctly reverse-engineers the technique, algorithm, or	Correctly identifies the technique/mechanism	Identification is only partially correct or lacks	Technique/mechanism is incorrectly identified or not addressed.	10

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
oning Mechanism	reasoning mechanism (e.g., chaining type, ML technique) with clear supporting evidence.	with adequate, mostly accurate justification.	sufficient supporting evidence.		
Quality of Justification & Use of Evidence (linking conclusions back to the given artifact)	Every conclusion is clearly justified with specific, accurate references to the given artifact.	Most conclusions are justified with reasonable references to the artifact.	Justification is vague or only loosely connected to the artifact.	Conclusions are stated with little or no justification or evidence.	10
Clarity & Completeness of Presentation (addresses all sub-parts of the question in an organised manner)	All sub-parts are addressed completely, in a clear, well-organised, and logically structured manner.	Most sub-parts are addressed clearly with minor gaps in organisation or completeness.	Some sub-parts are missing or the presentation is unclear/poorly organised.	Response is incomplete, disorganised, or largely missing.	10
				Total	50

Q1. AI-Based Flight-Booking Chatbot

(a) PEAS Description

PEAS Component	Description
Performance Measure	Correct flight search, understanding user requirements, relevant recommendations, low response time, successful booking, user satisfaction, accurate price and availability.
Environment	Users, airline databases, flight schedules, airport information, booking systems, payment systems and external APIs.
Actuators	Chat messages, flight recommendations, asking questions, displaying flight details, booking/cancelling flights and sending confirmation.
Sensors	User text messages, destination, date, time preference, airline preference, booking status and flight database/API responses.

PEAS in simple form: Performance = accurate and efficient flight booking; Environment = user + airline/flight database + booking services; Actuators = text responses, flight display and booking confirmation; Sensors = user input and flight/API information.

(b) Type of Agent

The chatbot is primarily a Model-Based Goal-Based Agent, with some characteristics of a Utility-Based Agent.

- It has a specific goal: help the user find and eventually book a suitable flight to Chennai.
- It maintains information about the conversation, such as destination, date, time and airline preference.
- It asks for missing information rather than reacting only to isolated keywords.
- The statement “Shall I show you the cheapest option first?” introduces utility-based behavior because price is considered when selecting an option.

Final classification: Model-Based Goal-Based Agent with Utility-Based Decision Support.

(c) Component Responsible for Each Response

A typical chatbot pipeline is: NLU → Dialogue Manager → Knowledge Base/Flight API → NLG.

Response 1: “Sure! What time would you like to depart, and do you have a preferred airline?”

- NLU extracts the intent (flight booking), destination (Chennai) and date (next Friday).
- The Dialogue Manager identifies missing slots: departure time and preferred airline.
- The NLG component converts the selected dialogue action into natural language.

Intent: BOOK_FLIGHT

Destination: Chennai

Date: Next Friday

Time: Unknown

Airline: Unknown

Response 2: “I found 3 morning flights to Chennai on Friday. Shall I show you the cheapest option first?”

- NLU extracts Time = Morning and Airline = No preference.
- The Dialogue Manager combines these with previously stored information.
- The Knowledge Base/Flight API returns three matching flights.
- The Dialogue Manager selects the next action, such as offering the cheapest flight first.
- NLG generates the final natural-language response.

Chatbot function	Likely component	Reason
Understanding Chennai/next Friday	NLU	Extracts intent and entities.
Asking for missing information	Dialogue Manager	Determines which slots are missing.
Finding three flights	Knowledge Base / Flight API	Retrieves availability.
Choosing cheapest option	Dialogue Manager + Knowledge Base	Applies user/system preference.
Generating the message	NLG	Converts decision into text.

(d) Enhancement Using a Learning-Agent Architecture

Add a Learning Element that learns user preferences from previous searches and bookings.

User → NLU → Decision/Action → Flight API

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Learning Element



Previous Bookings



Critic/Feedback

The system could learn that a user usually prefers morning flights, low prices, non-stop flights or particular airlines. After repeated bookings, the learned preference profile can be used to rank future recommendations.

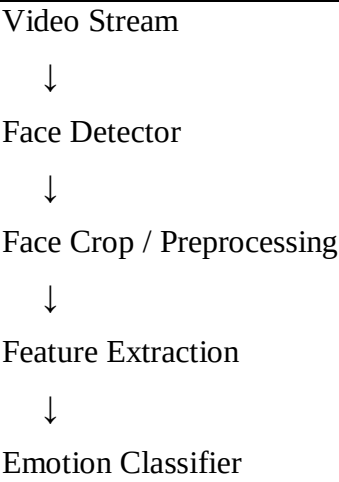
Benefit: the chatbot becomes personalized and adaptive instead of behaving exactly the same for every user.

Q2. Real-Time Emotion Detection System

Note: The actual block diagram was not included in the supplied question. The answer assumes a typical pipeline: Camera/Microphone → Preprocessing → Feature Extraction → Emotion Model → Emotion Classification → Output.

(a) Function of Each Stage

Stage	Function	Likely AI/ML technique
Input Acquisition	Captures real-time video frames and optionally audio.	Camera/microphone sensing
Face Detection	Locates the human face in each frame.	Haar Cascade, HOG, SSD, YOLO or MTCNN
Preprocessing	Crops, resizes, normalizes and optionally denoises the face image.	Image processing
Feature Extraction	Extracts facial structure, eyes, eyebrows, mouth and texture information.	CNN/deep embeddings, facial landmarks, HOG
Emotion Classification	Maps extracted features to emotions such as happy, sad, angry, fear, surprise, disgust and neutral.	CNN, transformer or classifier with softmax
Real-Time Output	Displays the predicted emotion or triggers an application action.	Decision/output logic



↓

Happy / Sad / Angry / Fear / Surprise / Disgust / Neutral

(b) Likely AI/ML Technique

A Convolutional Neural Network (CNN) is a likely technique because the primary input is facial image data. CNNs can learn edges, shapes, facial structures, eye patterns and mouth patterns.

Input Image → Convolution → ReLU → Pooling → Convolution → Pooling → Fully Connected → Softmax → Emotion

For sequential facial expressions, LSTM/GRU or a temporal transformer can be added after visual feature extraction.

(c) PEAS Description

PEAS	Description
Performance	High emotion-classification accuracy, low latency, stable predictions and correct detection.
Environment	People, lighting conditions, camera environment and background.
Actuators	Emotion display, dashboard update, notification/alert and application response.
Sensors	Camera, microphone, video frames and audio signals.

(d) Agent Type

The system can be classified as a Model-Based Reflex Agent when it maintains information from previous frames.

Frame 1 → Happy

Frame 2 → Happy

Frame 3 → Happy

Frame 4 → Happy

↓

Stable emotion = Happy

It is model-based when recent history is used to stabilize the prediction. If each frame is classified independently with no memory, it is closer to a Simple Reflex Agent.

Q3. Medical Expert System Shell

Rules: R1: fever = yes AND cough = yes $\rightarrow$ suspect = flu; R2: suspect = flu AND body_ache = yes $\rightarrow$ diagnosis = Influenza; R3: fever = yes AND rash = yes $\rightarrow$ suspect = measles.

Given facts: fever = yes, cough = yes, body_ache = yes.

fever + cough
↓
R1
↓
suspect = flu
↓
flu + body ache
↓
R2
↓
diagnosis = Influenza

(a) Type of Chaining

The system uses Forward Chaining because the reasoning is data-driven: it starts from known facts and repeatedly fires rules whose conditions are satisfied.

1. R1 fires because fever = yes and cough = yes.
2. R1 derives suspect = flu.
3. R2 fires because suspect = flu and body_ache = yes.
4. R2 derives diagnosis = Influenza.

Reasoning chain: Facts $\rightarrow$ R1 $\rightarrow$ suspect = flu $\rightarrow$ R2 $\rightarrow$ Influenza.

(b) Expert-System Components

Component	Role in this example
Knowledge Base	Stores medical IF–THEN rules R1, R2 and R3.
Inference Engine	Matches facts against rule conditions and fires R1 and then R2.
Explanation Facility	Generates the reasoning trace explaining why Influenza was concluded.

The given explanation — “Diagnosis = Influenza because fever and cough indicated flu, and flu together with body ache confirmed Influenza.” — is produced by the explanation facility.

(c) Knowledge Acquisition for a Dengue Rule

Knowledge acquisition obtains new knowledge from a domain expert and converts it into a formal rule.

5. Interview a medical expert about dengue symptoms and important diagnostic evidence.
6. Convert the expert's knowledge into a formal IF–THEN rule.
7. Validate the rule using sample cases.
8. Add the validated rule to the knowledge base.
9. Test for conflicts or unintended overlap with existing rules.

R4:

IF fever = yes

AND rash = yes

AND platelet_count = low

THEN diagnosis = Dengue

A test case such as fever = yes, rash = yes and platelet_count = low should produce diagnosis = Dengue.

(d) Could Backward Chaining Reach the Same Diagnosis?

Yes. Backward chaining is goal-driven. It begins with the goal diagnosis = Influenza and works backward to find rules that can establish that goal.

Goal: diagnosis = Influenza

↓

Find R2

↓

Need: suspect = flu AND body_ache = yes

↓

body_ache = yes (given)

↓

Need: suspect = flu

↓

Find R1

↓

Need: fever = yes AND cough = yes

↓

Both are given

↓

suspect = flu



Influenza

Forward Chaining	Backward Chaining
Starts with known facts	Starts with a goal
Data-driven	Goal-driven
Facts $\rightarrow$ conclusion	Goal $\rightarrow$ required facts
$R1 \rightarrow R2$	$R2 \rightarrow R1$
Reaches Influenza	Also reaches Influenza

Q4. Autonomous Drone for Disaster Detection with Edge AI

Note: The actual architecture diagram was not included in the supplied question. The answer assumes a typical Edge-AI disaster-detection drone with sensors/cameras, onboard AI, flight controller, obstacle avoidance, communication and cloud components.

(a) PEAS Description

PEAS	Description
Performance	Accurate disaster detection, safe navigation, obstacle avoidance, maximum coverage, low response time and efficient battery usage.
Environment	Disaster zones, buildings, roads, forests, smoke, fire, damaged infrastructure, debris, people and obstacles.
Actuators	Motors, propellers, flight-control system, camera gimbal, alarms and communication/transmission system.
Sensors	RGB camera, thermal camera, LiDAR/ToF, GPS, IMU, altitude sensor and proximity sensors.

Important performance measures include victim-detection accuracy, safe navigation, area coverage, battery efficiency and communication reliability.

(b) Goal-Based or Utility-Based Agent?

The drone is best classified as a Utility-Based Agent, although it also has goals.

- Goals include searching the disaster area, detecting victims, avoiding obstacles and reaching a destination.
- However, it must choose among alternative actions while considering safety, coverage, battery and communication.
- A utility function can combine detection accuracy, safety, area coverage, battery efficiency and communication quality.

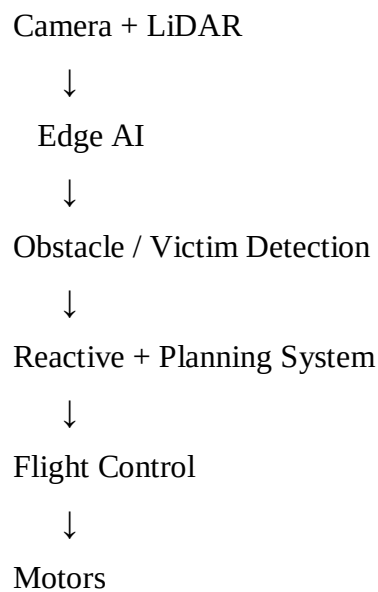
Utility = Detection Accuracy + Safety + Area Coverage + Battery Efficiency +
Communication Quality

Therefore, it is utility-based because it selects actions that maximize overall usefulness rather than merely checking whether a goal has been achieved.

(c) Edge vs Cloud Components

Edge / On-Drone	Cloud
Camera processing	Large-scale data storage
Object/victim/fire/smoke detection	Historical analytics
Obstacle detection	Model retraining
Flight control and stabilization	Central monitoring dashboard
Local path planning	Disaster mapping and reports
Emergency response logic	Multi-drone mission-level analytics

Edge AI is chosen for detection and obstacle avoidance because these functions are safety-critical and require very low latency. Disaster areas can also have weak or unavailable connectivity. Local inference allows the drone to continue operating without waiting for cloud responses and reduces bandwidth by transmitting important detections instead of continuous raw video.

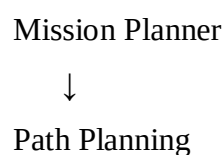


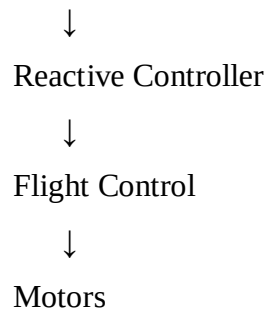
(d) Reactive, Deliberative, or Hybrid?

The drone uses a Hybrid Architecture because it combines immediate reactive control with higher-level deliberative planning.

Reactive layer: obstacle avoidance, emergency braking, altitude stabilization, collision prevention and immediate motor control.

Deliberative layer: mission planning, area coverage planning, route selection, waypoint generation and return-to-base decisions.





Final classification: Hybrid agent architecture — reactive behavior handles urgent safety decisions while deliberative planning handles mission-level objectives.

Q5. Deepfake & Face Authenticity Verification

Frame	Blink Rate (per 10s)	Texture/Artifact Score	Fake Confidence	Prediction
1	4	0.12	8%	Real
2	1	0.61	74%	Fake
3	0	0.83	91%	Fake
4	5	0.09	5%	Real
5	2	0.55	68%	Fake

Overall system verdict: Fake — 3 of 5 frames were flagged as Fake, so the system used majority voting.

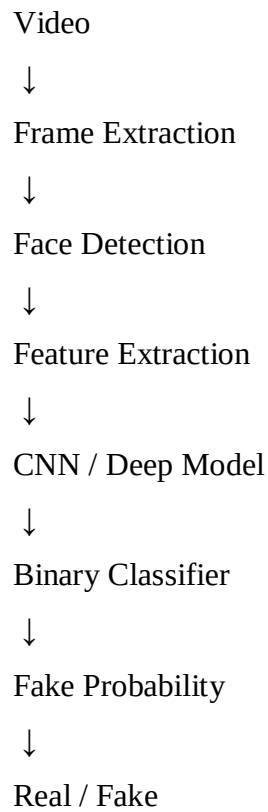
(a) Likely Input Features

Feature	What it captures
Blink rate	Frequency of blinking and possible abnormal temporal behavior.
Texture/artifact score	Unnatural skin texture, blending errors, warping, lighting inconsistencies and pixel-level artifacts.
Facial landmarks	Positions and relationships of eyes, nose, mouth, jaw and eyebrows.
Temporal consistency	Changes in facial motion, head pose, mouth movement and lighting across consecutive frames.
Audio-video synchronization	Consistency between spoken audio and lip movement, when audio is available.
Frequency-domain features	Unusual image-frequency patterns associated with generated or manipulated content.

The sample suggests that higher artifact scores and abnormal blinking are associated with higher fake confidence, but no single feature should be treated as sufficient evidence of manipulation.

(b) Likely Model/Technique

The system most likely uses a binary classifier with two classes: Real and Fake. A CNN is a likely visual model because it can learn spatial artifacts from face images.



For temporal information, CNN-LSTM, CNN-GRU or a transformer can be used. A sigmoid output can represent fake probability, for example $0.74 = 74\%$ fake confidence.

(c) Final Real/Fake Decision

Assuming a frame-level threshold of 50% fake confidence:

Frame	Fake Confidence	Frame Decision
1	8%	Real
2	74%	Fake
3	91%	Fake
4	5%	Real
5	68%	Fake

Counts: Real = 2 and Fake = 3. Because Fake has the majority, the overall system verdict is Fake.

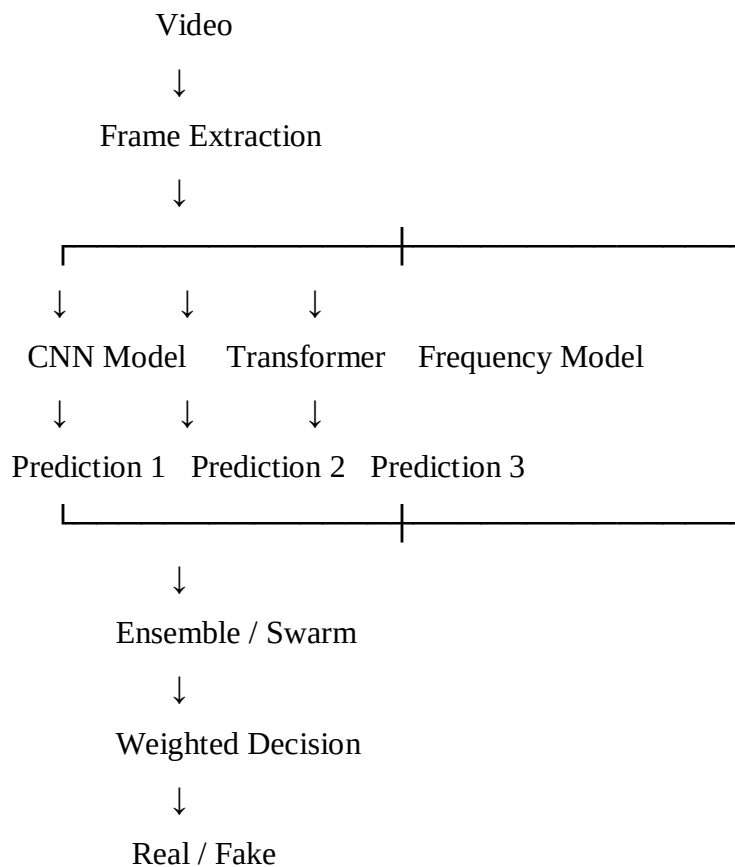
$\text{FinalDecision} = \text{Mode}(\text{Frame Predictions})$

$\text{Mode}(\text{Real, Fake, Fake, Real, Fake}) = \text{Fake}$

If average confidence were used instead, the result would be different: $(8 + 74 + 91 + 5 + 68) / 5 = 49.2\%$. Therefore, the question's stated verdict confirms that majority voting, rather than average confidence with a 50% threshold, is the final decision mechanism.

(d) Swarm/Ensemble-Based Improvement

A strong improvement is to combine several complementary deepfake detectors rather than relying on a single model.



Example: CNN = 82% Fake, Transformer = 91% Fake and Frequency model = 40% Fake. A weighted ensemble can combine these independent opinions. Models can specialize in spatial artifacts, temporal relationships, frequency abnormalities, facial geometry or audio manipulation.

- Reduces single-model bias.
- Combines different types of evidence.
- Improves robustness to new or sophisticated deepfake techniques.
- Allows confidence calibration and uncertainty handling.
- Can use model weights based on validation performance.

Final Reverse-Engineering Summary

Question	Main Finding
Q1	Flight chatbot is a model-based, goal-oriented agent with utility-based decision support.

Q1 PEAS	Performance = accurate booking; Environment = users/airlines; Actuators = responses/booking; Sensors = text/API information.
Q1 Enhancement	Learning element for personalized flight preferences.
Q2	Real-time emotion system uses camera → preprocessing → feature extraction → emotion classification.
Q2 Agent	Model-based reflex when previous frames are used; simple reflex if each frame is independent.
Q3	Medical expert system uses forward chaining; backward chaining can also reach the same diagnosis.
Q3 Components	Knowledge Base + Inference Engine + Explanation Facility; knowledge acquisition adds validated expert rules.
Q4	Drone is primarily utility-based.
Q4 Edge AI	Detection, obstacle avoidance and flight-critical processing run locally; cloud handles storage, analytics and retraining.
Q4 Control	Hybrid architecture: reactive obstacle avoidance + deliberative mission planning.
Q5	Likely features include blink rate, texture/artifacts, facial landmarks and temporal consistency.
Q5 Model	CNN/binary classifier; CNN-LSTM or transformer can model temporal behavior.
Q5 Decision	Majority voting: 3 Fake vs 2 Real → Fake.
Q5 Improvement	Ensemble/swarm of complementary deepfake detectors improves reliability.

Quick Exam Revision

Concept	Remember
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PEAS	Performance, Environment, Actuators, Sensors.
Simple Reflex	Current percept → immediate action.
Model-Based	Maintains internal state/history.
Goal-Based	Chooses actions to achieve a goal.
Utility-Based	Chooses action with best overall utility.
Learning Agent	Improves behavior using experience and feedback.
Knowledge Base	Stores domain rules and facts.
Inference Engine	Applies rules to facts.
Explanation Facility	Explains why a conclusion was reached.
Knowledge Acquisition	Adds and validates expert knowledge.
Forward Chaining	Facts → Rules → Conclusion.
Backward Chaining	Goal → Rules → Required Facts.
Edge AI	Low-latency, safety-critical local processing.
Cloud	Heavy computation, storage, analytics and training.
Hybrid Control	Reactive immediate response + deliberative planning.